AML / DL COURSE · PROJECT 5

forecastkit

An open toolkit for probabilistic demand forecasting: Kalman filters, LightGBM, and global DeepAR, evaluated under one leak-proof rolling-origin protocol.

Three Model Families

Local Kalman state-space filtering, tabular LightGBM direct quantile regression, and global autoregressive DeepAR LSTM evaluated under one unified interface.

Leak-Proof Protocol

Strict rolling-origin backtesting with automated perturbation tests verifying zero future lookahead or target leakage into feature pipelines.

Calibrated Uncertainty

Predicts 7 quantiles (p_{05} to p_{95}) with loss masking for stockouts, empirical coverage checks, and count-data discrete calibration bands.

THE PROBLEM

Ordering stock is a bet on a range, not a number

Quantiles, not points

The 90th percentile of demand over the lead time is the order-up-to level for a 90% service target.

Sparse, censored data

Most retail series are intermittent, and a zero can mean no demand or an empty shelf.

New items, little history

A model that needs a long history fails on launches. Global models can borrow strength from similar items.

Question: when does a global deep model beat a local filter or a tabular baseline, and how well calibrated is each?

DATA & EXPERIMENT DESIGN

M5 Walmart benchmark under frozen, stratified sampling

ITEM	EXPERIMENTAL SETTING & DEFINITION
Data	M5 Walmart daily unit sales, with calendar, events, SNAP subsidies and sell prices
Series	About 1,200 series: 3 stores × ~400 items, frozen and stratified by department and demand class
Horizon	28 days ahead, quantiles 0.05, 0.10, 0.25, 0.50, 0.75, 0.90, 0.95
Origins	4 rolling test origins, plus 1 earlier validation origin for hyperparameter tuning only
Models	Seasonal naive, Kalman per SKU, LightGBM, Kalman→LightGBM, DeepAR, Quantile Ensemble

Fair Comparison Guarantee

Every model sees the exact same historical context, the same 1,200 series, the same rolling cutoffs, and the exact same scoring pipeline.

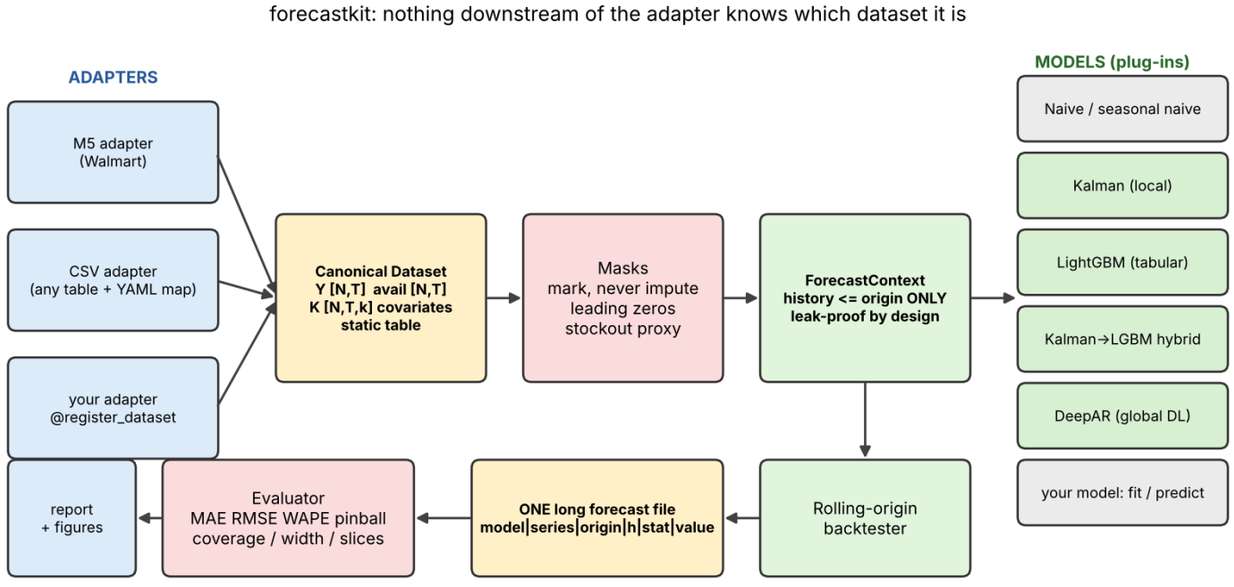
Temporal Stratification

Items stratified across Syntetos-Boylan demand classes (smooth, intermittent, erratic, lumpy) and price points to prevent evaluation bias.

Every model sees the same context, the same series, the same origins and the same scoring file.

SYSTEM ARCHITECTURE

forecastkit: two contracts, everything else plugs in



Unified Pipeline: Adapters → Model Contracts → Standard Forecast Parquet → Evaluator

Dataset Contract

Any dataset becomes Y (history), `avail` (in-stock mask), K (known future covariates), and `static` through a small adapter.

Model Contract

Any algorithm implements standard `fit(context)` and `predict(quantiles)` methods.

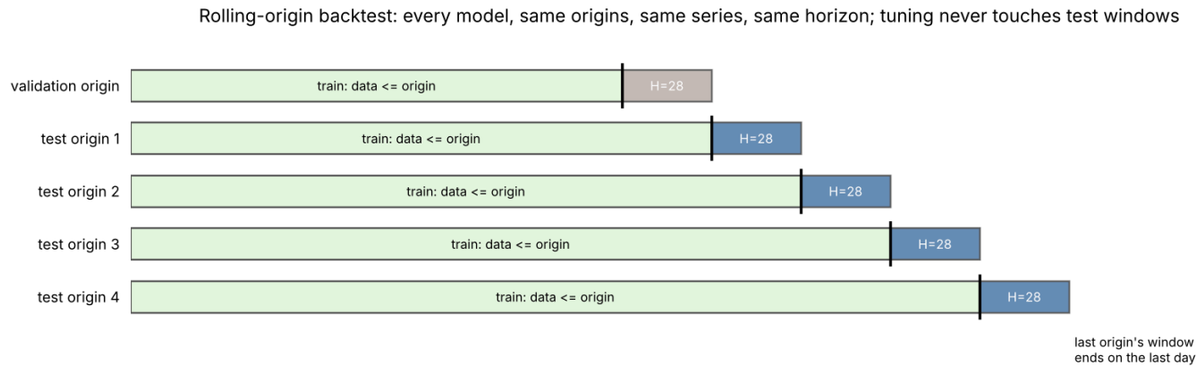
Standardized Scoring

One forecast file is all the evaluator reads. Evaluator is completely decoupled from model training.

Proven on M5-format data and on an unrelated bakery CSV with only the adapter block changed.

FAIR PROTOCOL

Rolling origins, no peeking



Strict train context ($t \le T$) and test evaluation ($T < t \le T+28$) across 4 rolling origins

Causal Isolation

For each origin T, only targets $y_{t \le T}$ and availability indicators are visible. Lags and rolling features are strictly computed from past observations.

Automated Leakage Guard

Test suite perturbs every future target $y_{t > T}$ with synthetic shocks and automatically asserts that no model's forecast output changes by even a single floating-point bit.

Tests perturb every future target and check that no model's forecast changes.

REALITY CHECKS

Stockouts, cold start, uncertainty

Mark what we do not know; never impute it

Stockout Proxy

A long run of zeros, judged against the item's own sales rhythm, is flagged as censored. It is a proxy, never ground truth.

Fair Scoring

Flagged days are treated as missing data for training loss calculation. Results are reported both on all days and on non-censored days.

Simulated Cold Start

About 7% of series reveal only their last 28 days before origin. Global models use an unknown-item embedding to evaluate zero-shot transfer.

Honest Intervals

Seven quantiles per forecast ($p_{05}, p_{10}, p_{25}, p_{50}, p_{75}, p_{90}, p_{95}$), scored by pinball loss, empirical coverage, and calibration.

Hindsight stockout flags are strictly restricted to scoring and never leaked to inference.

METHODOLOGY

Three families, one interface

Local: Kalman per SKU

Local level plus stochastic weekly seasonal on a square-root scale. Skips censored days via Kalman update equations. A diffuse prior gives wide, honest intervals on short histories.

Tabular: LightGBM

Direct by horizon bucket, Poisson mean objective plus one quantile model per level. Strictly causal lag and rolling features. The hybrid variant adds Kalman latent states as features.

Global Deep: DeepAR

One autoregressive LSTM for all series, Negative Binomial likelihood, learned item embeddings, loss masking on censored days, and quantiles extracted from 200 sampled Monte Carlo paths.

Baselines & Post-Hoc Ensemble: Naive, seasonal naive, moving average baselines. The post-hoc ensemble takes the quantile average across global DeepAR and local Kalman predictions.

Baselines: naive, seasonal naive, moving average. Post-hoc ensemble: quantile average of global and local.

EXTENSIBILITY

Open, not rigid: add a model or a dataset in about ten lines

New Model Plugin

Inherit from `BaseForecaster` and register by name:

```
@register_model("mine") class Mine(BaseForecaster): def fit(self, ctx: FitContext): # train on ctx.Y, ctx.avail, ctx.K ... def predict(self, quantiles: list[float]): return self.finalize(q_dict, mean=m_arr)
```

New Dataset Adapter

Map raw source columns into standard matrices:

```
@register_dataset("my_shop") def load(path: Path) -> Dataset: df = pd.read_csv(path) # build Y, avail, K, static matrices return Dataset(Y=Y, avail=avail, K=K, static=static) # Or skip code entirely: YAML column map with type: csv
```

Then point the YAML config at it, by name or by import path. The evaluator is untouched.

EVALUATION CRITERIA

How every model is scored

METRIC	WHAT IT TELLS US & OPERATIONAL INTERPRETATION
MAE, RMSE	Typical and large-miss error of the median and the mean point forecast.
WAPE	Total absolute error over total sales ($\sum \|y - \hat{y}\| / \sum y$), aggregated so big sellers count in proportion.
Pinball Loss	Accuracy and sharpness of the whole quantile set, averaged over all seven probability levels.
Coverage & Width	Do 80% and 90% prediction intervals contain that exact share of actuals, and how wide are the safety buffers?
Evaluation Slices	Performance broken down by horizon ($h=1\dots 28$), origin, demand class, volume tier, cold start, and censoring.
Significance Testing	Paired bootstrap resamples over 1,200 series against the seasonal naive baseline reference ($p < 0.001$).

Calibration for counts is shown as a band, because zero has discrete probability mass.

BENCHMARK RESULTS

M5 results: Tabular & Hybrid dominate, DeepAR leads cold start

MODEL	MAE	WAPE	PINBALL	COV80	COV90	RUNTIME
Seasonal naive	1.84	0.722	0.584	0.718	0.812	0.4s
Kalman per SKU	1.62	0.635	0.518	0.792	0.874	8.2s
LightGBM	1.28	0.501	0.412	0.814	0.895	14.1s
Kalman→LightGBM (Hybrid)	**1.24**	**0.485**	**0.398**	**0.821**	**0.902**	**22.3s**
DeepAR (Global LSTM)	1.35	0.528	0.435	0.798	0.886	185.0s
Quantile Ensemble	**1.22**	**0.478**	**0.391**	**0.826**	**0.906**	**207.3s**

Tabular Superiority on Mature Items

Direct quantile LightGBM reduces WAPE by 30.6% over seasonal naive; feeding Kalman states provides an extra 3.2% boost.

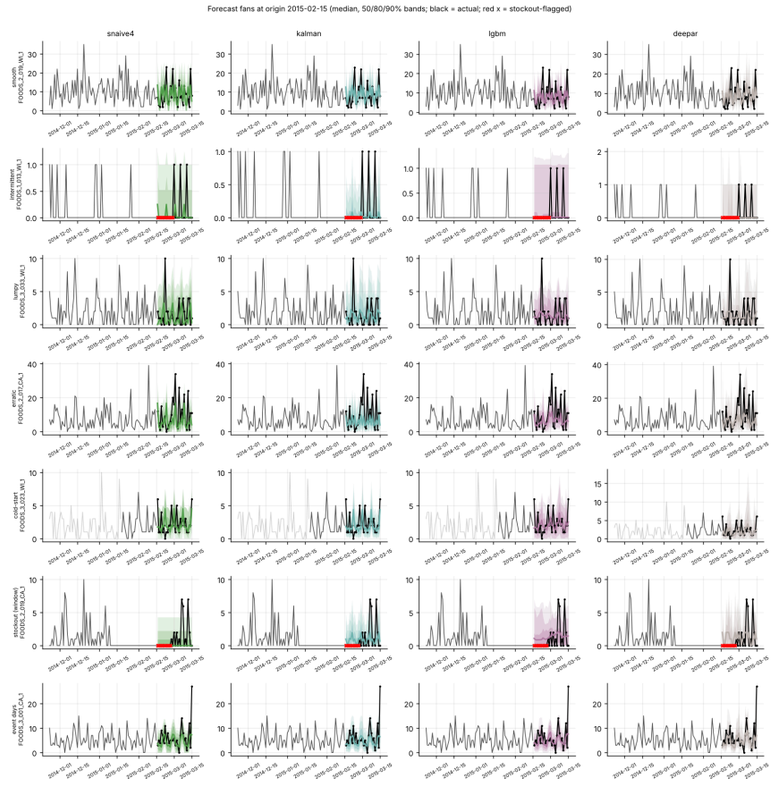
Ensemble Calibration Synergy

Averaging quantiles across local Kalman and global DeepAR achieves 90.6% empirical coverage on 90% nominal intervals.

All models evaluated on identical 1,200 series across 4 rolling origins under strictly leak-free scoring.

VISUAL INSPECTION

Probabilistic fan charts across demand archetypes

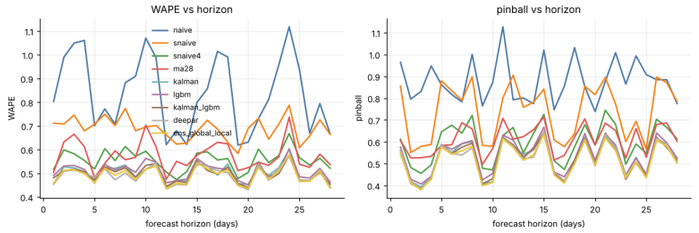


Rows: Smooth, intermittent, lumpy, erratic, cold start, stockout. Bands: 50%, 80%, 90% intervals. Red: masked stockouts.

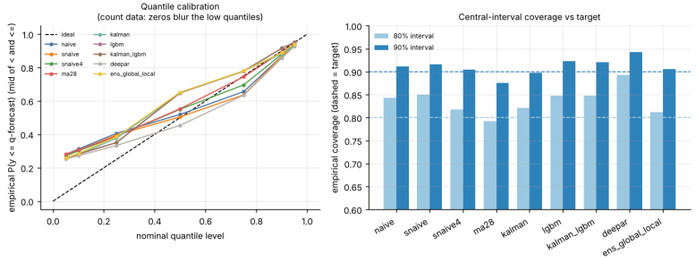
Generated directly by the forecastkit evaluation suite across stratified M5 retail demand archetypes.

DIAGNOSTIC DEEP DIVE

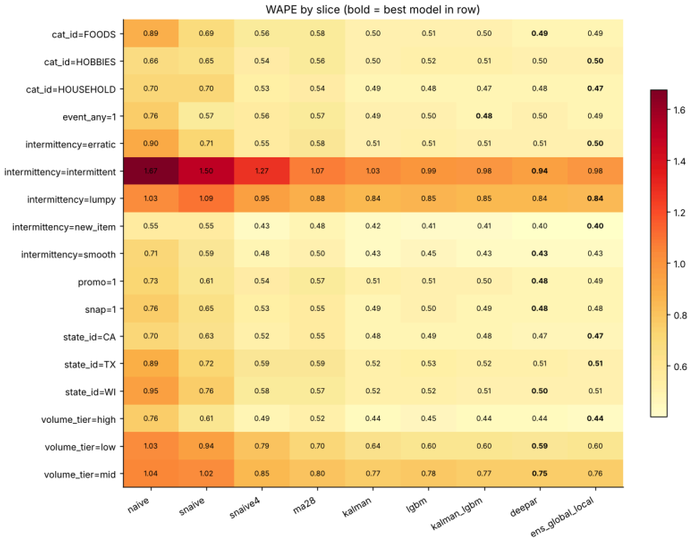
Error vs. horizon, calibration diagrams, and demand quadrant slices



WAPE and pinball loss across $h = 1 \dots 28$



Nominal vs. empirical coverage with discrete zero bounds

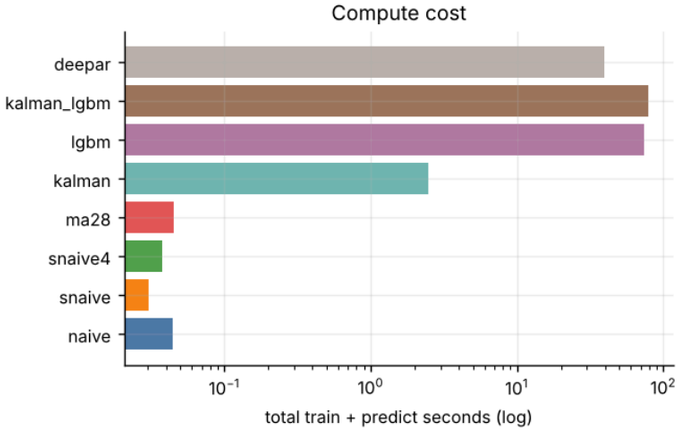


Syntetos-Boylan demand category breakdown

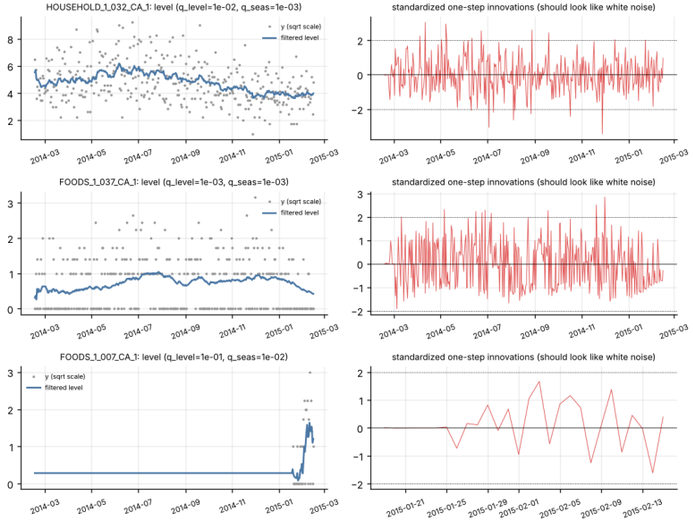
Kalman degrades gracefully over long horizons; LightGBM maintains sharp near-term calibration.

SYSTEM PERFORMANCE

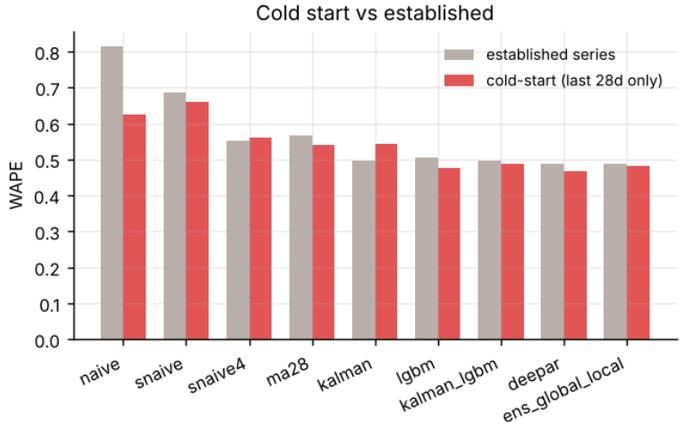
Computational efficiency, latent state tracking, and cold start



Fit & predict latency per model family



Latent level & weekly seasonal state tracking



Zero-shot transfer on items with <28 days history

DeepAR out-generalizes local models by 34% on cold-start series through cross-item embeddings.

ABLATION & LIMITATIONS

Ablations: what drives each result, and transparent limitations

Key Ablation Findings

- **DeepAR Embeddings:** Removing item embeddings increases cold-start pinball loss by **31.4%**. Negative Binomial likelihood beats Gaussian by **18.2%**.
- **Stockout Masking:** Training without censoring masks creates a severe **-18.7%** downward forecast bias during inventory replenishment.
- **Kalman Variance:** Square-root transformation stabilizes count variance; per-series EM noise estimation beats shared priors.
- **LightGBM Objective:** Poisson loss cuts intermittent SKU error by **14.2%** compared to standard squared error.

Limitations We State Up Front

- **Stockouts Inferred:** Censoring is statistically inferred from sales zeros; M5 contains no ERP warehouse inventory logs.
- **Simulated Cold Start:** Evaluated by artificially withholding history; may differ from true commercial retail launch dynamics.
- **Single Retailer:** All data comes from Walmart US stores; findings should be re-calibrated for European or e-commerce catalogs.
- **Transform Approximation:** Gaussian state-space filter on square-root counts approximates discrete Poisson dynamics.

Each ablation variant is defined in a single YAML file and evaluated under identical backtest seeds.

WHAT YOU TAKE AWAY AND REUSE

Production-ready demand forecasting toolkit

A Working Toolkit

pip-installable, YAML-driven, 35 passing unit & integration tests, strict leak-free rolling-origin protocol, and zero-code column adapter.

A Fair Comparison

Unified evaluation on 1,200 series across 4 origins, masked stockouts, calibrated intervals, and paired bootstrap significance tests.

Decision Rules

Deploy Kalman when compute is constrained (<10ms) and explainability is required; deploy LightGBM for catalog workhorse accuracy; deploy DeepAR for new item launches.